Raj Khatri

Embedded Firmware Engineer

RESUME_1719

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Total Experience: 10 years

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 10 years of experience in embedded firmware, using C, C++, ARM Cortex-M, FreeRTOS. Experienced in real-time systems, hardware debugging and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: C++, Firmware Debugging, Embedded Linux, FreeRTOS, STM32, Git, ARM Cortex-M, C, Oscilloscope, SPI, UART, I2C

Methods: Real-time systems, Hardware debugging, Protocol validation

PROFESSIONAL EXPERIENCE

Lead Embedded Firmware Engineer | Foresight Industrial | Jan 2023 - Present

- Lowered field-reset incidents by 24% with watchdog recovery and diagnostics
- Reduced embedded task jitter by 26% using interrupt priority and queue tuning
- Improved battery runtime by 18% with peripheral and sleep-state optimization
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Embedded Firmware Engineer | DeltaPoint Innovations | Jan 2019 - Dec 2022

- Improved battery runtime by 15% with peripheral and sleep-state optimization
- Validated 144 hardware test vectors across protocol and timing corner cases

Associate Embedded Firmware Engineer | Ironleaf Engineering | Jul 2016 - Dec 2018

- Lowered field-reset incidents by 20% with watchdog recovery and diagnostics
- Reduced embedded task jitter by 17% using interrupt priority and queue tuning

PROJECTS

Low-Power Metering Device | STM32, ARM Cortex-M, I2C, Git

- Optimized interrupt-driven acquisition and sleep cycles.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

Bootloader Update Utility | C, ARM Cortex-M, STM32, FreeRTOS

- Developed verified firmware transfer and rollback checks.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

EDUCATION

B.Tech in Electronics and Communication Engineering, Crescent Valley Technical University, 2016

Relevant Methods: Real-time systems, Hardware debugging, Protocol validation

CERTIFICATIONS

RTOS Scheduling Training (2019)